Samarth Batra

New York, NY | samarth.batra@stonybrook.edu | Linkedin | Github | Website

Profile

Software Engineer with 3+ years of experience building distributed backend systems and cloud-native applications using Java, Spring Boot, and AWS. Skilled in designing scalable APIs and data pipelines, with proven success at Amazon and Verint.

Education

Stony Brook University — **Aug 2024 – Dec 2025**
Master's in Computer Science — *Stony Brook, NY*
Course Work : Distributed Systems | Database Systems | Machine Learning | Computer Architecture

University of Petroleum and Energy Studies — **Jul 2017 – May 2021**
Bachelor of Science in Computer Science and Engineering — *Dehradun, UK*
Course Work : Data Structures & Algorithms | Operating Systems | Database Management | Object-Oriented Programming

Technical Skills

Programming Languages: Java, Python, JavaScript, C/C++, TypeScript, SQL, PL/SQL, HTML/CSS, Groovy, Shell Scripting
Frameworks: Spring Boot, Angular, React.js, Redux, Babel, TailwindCSS, BootStrap, Node.js, Django, Express.js, Webpack, Junit
Databases: Oracle, MongoDB, MySQL, Microsoft SQL Server, PostgreSQL
CI & CD: Jenkins, Docker, Kubernetes, Maven, Git, GitHub, Terraform
Cloud Technologies: Microsoft Azure, Google Cloud Platform (GCP), AWS, Azure HDInsight, Azure DevOps, Azure DataLake
Tools & Technologies: ELK Stack, Redis, RabbitMQ, gRPC, Spark, Hadoop, Swagger, Jmeter, Airflow, NLTK, GPT

Experience (3+ years of professional experience)

Verint Systems — **May 2025 – Aug 2025**
*Software Developer Research Intern — **Java, Spring Boot, JPA, Gradle, GitHub Actions*** — *New York, NY*

- Designed and implemented the **DAO and persistence layer** for the Platform Admin Service (core administrative service for tenant and experience management), including **JPA entities, repositories, converters, and transactional flows**.
- Developed a **DTO–Entity converter layer** for Experience Selector and Condition models, ensuring **clean separation of concerns, consistency across the service**, and **reduction of boilerplate code**.
- Integrated **JaCoCo coverage** into the Gradle build and GitHub Actions workflows to automatically **post PR comments with detailed coverage metrics**, including total code coverage and per-module breakdown.

Amazon — **May 2022 – Aug 2024**
*Software Developer | **Python, Java, TypeScript, REST APIs, React, AWS*** — *Bangalore, KA*

- Designed and developed 10+ high-level (**HLD**) and low-level designs (**LLD**) for **test account separation**, **product detail pages**, and **real-time inventory management**, improving system modularity and deployment efficiency.
- Architected and deployed low-level API solutions for reimbursement event reporting, introducing **ListReimbursements** and **ListReimbursementsByOrderId** APIs to enable granular filtering, reducing query response times by **2 seconds**.
- Implemented **global secondary indexes** in **DynamoDB** to optimize query performance and reduce latency for reimbursement event retrieval by **20%**, enabling faster data access and improved user experience.
- Adopted **Transitive Auth (TA)** and **CloudAuth** for **CARESCoreService**, implementing fine-grained access controls, secure context propagation, and **TA-Token integration** to enhance customer data security and system reliability.
- Created and incorporated data models and methods within **SCPBillingService**, **SCPBillingServiceModel**, and **SupplyChainPortalMonsWrapperService**, optimizing data flow and improving system efficiency.
- Enhanced UI functionality by adding dropdown filters for **Order Id**, **Statement Id**, and **Transaction Id** using **React.js**, integrating them with the backend **ListReimbursements API** I **engineered**, improving user experience and search flexibility.

Cognizant Technology Solutions — **Jan 2021 – May 2022**
*Software Engineer | **Java, TypeScript, Angular, Python, Webpack, Azure, LLMs*** — *Bangalore, KA*

- Created and delivered an **interactive dashboard** for real-time data visualization using **D3.js** and **Angular**, leveraging **NgRx** for state management, implementing **lazy loading** for improved performance and **Axios** for API communication.
- Optimized front-end performance by implementing **lazy loading**, code splitting, and **browser caching**, reducing page load time by **40%** and increasing site engagement.
- Optimized application performance using **Webpack** and **Babel**, reducing the application's load time by 35% and increasing Core Web Vitals scores.

Projects & Research

Web Control with Eye Tracking and Voice Commands (Chrome Web Extension) link
Skills : ***Javascript, LLMs, GPT, BERT, NLTK, spaCy, Google Cloud Platform (GCP), NLP.*** — **Aug 2024 – Dec 2024**

- Engineered a **Chrome browser extension** integrating **WebGazer.js** for real-time eye-tracking (95% accuracy) and **voice recognition** using NLP and DOM interaction, enabling hands-free scrolling, clicking, and navigation with visual feedback, reducing interaction time by **40%** and ensuring precise, scalable web control.

Small-Scale Hypertextual Web Search Engine
Skills : ***Python, Flask, MongoDB, BootStrap, AWS.*** — **May 2022**

- Built a web crawler in **Python** to index and store data in **MongoDB**, processing 10,000+ pages, and developed a ranking algorithm to improve search efficiency by **30%** and reduce query response time by **20%**. Designed a user-friendly **Flask** interface, increasing user engagement by **25%** and enhancing overall user experience. .